Common Packages in R and Python

A categorized reference of packages with overlapping functionality. These are practical counterparts, not always exact equivalents.

Reading the tables: Package names are shown as plain text; base R and Python standard-library tools are identified where relevant. Availability and APIs can change over time.

1. Statistics and probability

Purpose	R package	Python package
Basic statistical tests	`stats` (base R)	`scipy.stats`
Statistical models	`stats` (base R)	`statsmodels`
Generalized linear models	`glm` (base R)	`statsmodels`
Bayesian statistics	`rstan`	`PyStan`, `CmdStanPy`
Bayesian probabilistic programming	`brms`	`PyMC`
Probability distributions and random variables	`stats` (base R)	`scipy.stats`
Survival analysis	`survival`	`lifelines`
Mixed-effects models	`lme4`	`statsmodels`
Time-series analysis	`forecast`, `fable`	`statsmodels`, `sktime`
Bootstrapping	`boot`	`scipy`, `arch`
Multiple imputation	`mice`	`miceforest`, `statsmodels`
Structural equation modelling	`lavaan`	`semopy`

2. Data manipulation and tabular data

Purpose	R package	Python package
Data frames	`data.frame` (base R)	`pandas`
Data manipulation	`dplyr`	`pandas`
Fast data manipulation	`data.table`	`polars`
Reshaping data	`tidyr`	`pandas`
Reading CSV files	`readr`	`pandas`
Reading Excel files	`readxl`	`openpyxl`, `pandas`
Working with large datasets	`data.table`, `arrow`	`polars`, `pyarrow`
Database access	`DBI`, `RSQLite`	`SQLAlchemy`, `sqlite3`
Data validation	`validate`	`pandera`
Missing-data analysis	`naniar`, `VIM`	`missingno`, `scikit-learn`

3. Data visualization and graphics

Purpose	R package	Python package
Grammar of graphics	`ggplot2`	`plotnine`
Statistical graphics	`lattice`	`seaborn`
Interactive plots	`plotly`	`plotly`
Interactive statistical visualization	`ggvis`	`altair`
Interactive web applications	`shiny`	`shiny` (for Python)
Interactive maps	`leaflet`	`folium`
Spatial data analysis	`sf`	`geopandas`
Interactive widgets	`htmlwidgets`	`ipywidgets`
3D visualization	`rgl`	`plotly`, `pyvista`
Graph and diagram visualization	`DiagrammeR`	`graphviz`, `networkx`

4. Machine learning and artificial intelligence

Purpose	R package	Python package
General machine learning	`caret`	`scikit-learn`
Machine-learning workflows	`tidymodels`	`scikit-learn`
Machine-learning framework	`mlr3`	`scikit-learn`
Random forests	`randomForest`	`scikit-learn`
Decision trees	`rpart`	`scikit-learn`
Gradient boosting	`xgboost`	`xgboost`
Gradient boosting	`lightgbm`	`lightgbm`
SVMs and related methods	`e1071`	`scikit-learn`
Neural networks	`nnet`	`scikit-learn`, `PyTorch`
Deep learning	`keras`	`keras`
Deep learning	`torch`	`torch` (PyTorch)
Deep learning	`tensorflow`	`tensorflow`
Regularized regression	`glmnet`	`scikit-learn`
Fast random forests	`ranger`	`scikit-learn`
Cluster analysis	`cluster`	`scikit-learn`

5. Numerical computing and mathematics

Purpose	R package	Python package
Sparse matrices	`Matrix`	`scipy.sparse`
High-performance computation	`Rcpp`	`Numba`, `Cython`
Linear algebra	`RcppArmadillo`	`NumPy`, `SciPy`
Numerical mathematics	`pracma`	`scipy`
Differential equations	`deSolve`	`scipy.integrate`
Root finding	`rootSolve`	`scipy.optimize`
Optimization	`optimx`	`scipy.optimize`
Computational geometry	`geometry`	`scipy.spatial`
Graph algorithms	`igraph`	`igraph`
Matrix exponentials	`expm`	`scipy.linalg`
Arbitrary-precision arithmetic	`Rmpfr`	`mpmath`
Arbitrary-precision integers	`gmp`	`gmpy2`
Symbolic mathematics	`Ryacas`	`SymPy`
Symbolic computation	`rSymPy`	`SymPy`

6. Econometrics and finance

Purpose	R package	Python package
Time-series forecasting	`forecast`	`statsmodels`
Vector autoregression	`vars`	`statsmodels`
Unit-root and cointegration tests	`urca`	`arch`, `statsmodels`
Panel-data econometrics	`plm`	`linearmodels`
Robust covariance estimation	`sandwich`	`statsmodels`
Regression diagnostics and tests	`lmtest`	`statsmodels`
Financial data analysis	`quantmod`	`yfinance`, `pandas`
Portfolio performance	`PerformanceAnalytics`	`empyrical`
GARCH models	`rugarch`	`arch`
Portfolio optimization	`PortfolioAnalytics`	`PyPortfolioOpt`
Technical indicators	`TTR`	`ta`, `pandas-ta`
High-dimensional econometric models	`fixest`	`pyfixest`

7. Text mining and natural language processing

Purpose	R package	Python package
Text mining	`tm`	`NLTK`
Text analysis	`quanteda`	`spaCy`, `NLTK`
Tidy text processing	`tidytext`	`scikit-learn`
String processing	`stringr`	`re` (standard library), `regex`
Text vectorization and embeddings	`text2vec`	`Gensim`
Topic modeling	`topicmodels`	`Gensim`
Sentiment analysis	`sentimentr`	`VADER`, `TextBlob`
Natural language processing	`spacyr`	`spaCy`
Text tokenization	`tokenizers`	Hugging Face Tokenizers

8. Bioinformatics and scientific research

Purpose	R package	Python package
Biological sequence analysis	`Biostrings`	`Biopython`
Genomic interval operations	`GenomicRanges`	`pyranges`
Differential gene expression	`DESeq2`	`PyDESeq2`
RNA-seq analysis	`edgeR`	`PyDESeq2`, `scanpy`
Experimental-data analysis	`limma`	`statsmodels`
Microbiome analysis	`phyloseq`	`scikit-bio`
Single-cell analysis	`Seurat`	`Scanpy`
Flow cytometry	`flowCore`	`FlowCytometryTools`
Raster data processing	`raster`	`rasterio`
Geospatial raster analysis	`terra`	`rasterio`, `xarray`

9. Web scraping, APIs, and applications

Purpose	R package	Python package
HTTP requests and APIs	`httr2`	`requests`, `httpx`
Web scraping	`rvest`	`BeautifulSoup`
Browser automation	`RSelenium`	`Selenium`
Creating APIs	`plumber`	`FastAPI`, `Flask`
Interactive applications	`shiny`	`Streamlit`, `Dash`
Computational pipelines	`targets`	`Prefect`, `Snakemake`
Unit testing	`testthat`	`pytest`
Project dependency management	`renv`	`venv`, `poetry`
R/Python interoperability	`reticulate`	`rpy2`
Reproducible reports	`knitr`	`Jupyter`, `nbconvert`
Reports combining code and prose	`rmarkdown`	`Quarto`, `Jupyter`

10. Same or closely related package projects in both ecosystems

Purpose	R package	Python package
Interactive plotting	`plotly`	`plotly`
Gradient boosting	`xgboost`	`xgboost`
Gradient boosting	`lightgbm`	`lightgbm`
Deep learning	`keras`	`keras`
Deep learning	`tensorflow`	`tensorflow`
Deep learning	`torch`	`torch` (PyTorch)
Graph analysis	`igraph`	`igraph`
Columnar data / Arrow	`arrow`	`pyarrow`
Analytical database	`duckdb`	`duckdb`
R/Python bridge	`reticulate`	`rpy2`

Important qualification

There is no single definitive list of every package common to R and Python: both ecosystems contain thousands of packages, and many overlap only in purpose. For official package repositories, consult CRAN (cran.r-project.org) and PyPI (pypi.org).